

# The economics behind the proofs: a short guide to the asymptotic knapsack reversal

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*The author takes sole responsibility for all claims.*

**Abstract.** This companion document provides plain-language economic intuition for the asymptotic dominance theorems in [Weinhold and Andersen \(2026\)](#). Both theorems establish that in adversarial procurement under a two-value structure, a simple value-maximisation rule (*value-first*, or ME) outperforms the textbook cost-effectiveness heuristic (*ratio-greedy*, or MX). The reversal has a single-sentence economic reading: *ratio-greedy retreats; value-first races; the race is worth the cost*. Under independence, this is captured exactly by a symmetry of the fluid-limit game dynamics, yielding the closed-form asymptotic gap  $\mu_E \phi - S_G(\tau) \approx 0.216$ . Under positive correlation the symmetry breaks; the gap is established instead by tracking the game's actual trajectory through a deterministic ODE closed with validated numerical integration (certified margin  $\geq 0.214$ ). A single economic mechanism — *self-hedging* — organises both results: the baseline symmetry is self-hedging in its purest form, and under positive correlation self-hedging decomposes into four distinct levels that together rescue the reversal. The simulated gap is hump-shaped in  $\rho$ , peaking near  $\rho \approx 0.7$ ; we show how two opposing forces — rising selection amplification and rising cross-damage — generate this profile. This document develops the full reading, explains why ecological heterogeneity is a necessary condition, and closes with a note on proof robustness: the baseline theorem has been established through three substantially different proof architectures, all converging on the same conclusion.

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# 1 The big picture

Weinhold and Andersen (2026) show that in an adversarial land market where a conservation agency (Green) competes against agricultural developers (Farmer) for sites drawn from a landscape of heterogeneous ecological and agricultural values, a simple “buy the best ecology” rule asymptotically outperforms the familiar bang-for-the-buck benchmark that underlies much conservation prioritisation. The result is surprising because the bang-for-the-buck rule (MX) embodies the fractional-knapsack intuition that is optimal when there is no competition. The theorems show that competition reverses this logic: when your rival’s purchases determine your costs, cheapness stops being an unambiguous virtue.

Two correlation values between ecological and agricultural attributes are formally covered, with different proof strategies:

- $\rho = 0$  (**independence**). An exact symmetry of the fluid-limit dynamics collapses the problem to a closed-form identity. The asymptotic gap is  $\mu_E \phi - S_G(\tau) \approx 0.216135$ , certified to five decimal places.
- $\rho = 0.3$  (**positive correlation**). The symmetry breaks. The proof works instead by tracking Green’s actual per-turn ecological value through a deterministic ODE on the survivor pool state, closed with validated numerical integration. The certified asymptotic gap is  $\geq 0.214$ .

Two proofs, one economic mechanism. Both routes terminate at the same economic conclusion — value-first wins because ratio-greedy selects into a corridor of the landscape that the rival does not contest — and they illuminate different facets of why that conclusion holds.

The rest of this document develops the mechanism. Section 2 states the economic idea (*self-hedging*) in its most general form. Section 3 shows how it collapses to a single symmetry identity at  $\rho = 0$ . Section 4 shows what changes when symmetry breaks, and why the reversal nevertheless survives at moderate positive correlation. Section 5 discusses the necessary condition — ecological heterogeneity — and the diagnostic that operationalises it for real landscapes. Section 6 closes with what the proofs do and do not show, including a note on the alternative

proof architectures that have now been completed for the baseline theorem.

## Numerical ledger

The reader may find it useful to anchor the narrative that follows to the key numerical quantities that appear in the proofs. All constants refer to the baseline DGP (uniform marginals on  $[0.1, 10]$ , Gaussian copula, budget parity) under the stated correlation.

| Symbol                                                    | Value                      | Meaning                                      | Where      |
|-----------------------------------------------------------|----------------------------|----------------------------------------------|------------|
| <i>Baseline</i> ( $\rho = 0$ )                            |                            |                                              |            |
| $\ell, \bar{u}$                                           | 0.1, 10                    | Marginal support                             | throughout |
| $\mu_E, \mu_A$                                            | 5.05                       | Marginal means                               | throughout |
| $\phi$                                                    | 0.295816                   | MX Farmer-exhaustion share                   | §3         |
| $\tau$                                                    | 0.304718                   | ME Farmer-exhaustion share                   | §3         |
| $\kappa = 1 - \tau$                                       | 0.695282                   | Green's exhaustion share                     | §3         |
| $\mu_E \phi$                                              | 1.493871                   | MX's cession by design                       | §3         |
| $S_G(\tau)$                                               | 1.277736                   | ME's racing cost                             | §3         |
| $\mu_E \phi - S_G(\tau)$                                  | <b>0.216135</b>            | <b>Baseline asymptotic gap</b>               | Thm. 4.1   |
| <i>Correlated</i> ( $\rho = 0.3$ )                        |                            |                                              |            |
| $m, \underline{M}_E(0.10)$                                | 0.10, 0.901                | Prefix share and ME prefix value lower bound | §4         |
| $\kappa_{\text{low}}^{\text{FA}}$                         | 0.620816                   | Certified Green affordability horizon        | §4         |
| $c_{\text{up}}^{\text{FA}}, c_{\text{up}}^{\text{total}}$ | $\leq 1.4723, \leq 2.5148$ | FA band cost and full certified cost         | §4         |
| $\int_m^{\kappa_{\text{low}}} e(c(s)) ds$                 | $\geq 2.5785$              | Fluid-limit ME ecological value              | §4         |
| $\mathbb{E}[E \mathbf{1}\{A \leq q_F\}]$                  | 3.265118                   | MX total asymptotic value                    | §4         |
| $\Theta_{0.3}$                                            | $\leq 2.36412$             | Outside-prefix MX benchmark                  | §4         |
| Certified margin                                          | $\geq$ <b>0.214</b>        | <b>Correlated asymptotic gap</b>             | Thm. 4.2   |
| <i>Diagnostics</i>                                        |                            |                                              |            |
| ESP (baseline, Bolivia)                                   | $\sim 0.22, > 0.30$        | Ecological selection premium                 | §5         |
| ESP threshold                                             | $\sim 0.15$                | Minimum for formal proof                     | §5         |
| $\rho$ -gap peak                                          | $\rho \approx 0.7$         | Location of hump maximum                     | §4         |
| Bolivia correlation                                       | $\rho \approx 0.29$        | Estimated latent correlation                 | §4         |

## 2 The mechanism: self-hedging

The central economic idea is that *the naïve structure of the Farmer's strategy hedges Green against the costs of its own success*. This is the self-hedging mechanism. It was hypothesised as the economic story behind the knapsack reversal early in the project, before any formal theorem was proved, and subsequent proofs have confirmed it as a mathematical property of the game rather than a post-hoc reading of simulation patterns.

Self-hedging has four distinct levels. Under positive correlation all four are active and jointly drive the reversal; at  $\rho = 0$  they collapse into a single clean symmetry, which is why the baseline

case admits the remarkably tidy proof that it does.

**Level 1 — value self-hedging.** When the Farmer removes the most expensive items from the market, those items are (under positive correlation) also ecologically valuable. Value-first has already secured some of the ecological elite in its opening moves; ratio-greedy was relying on those items being available later at a good ratio. Farmer’s removals systematically destroy ratio-greedy’s later options while leaving value-first’s portfolio intact. The value gap between the two opening portfolios — the *prefix wedge* — is the engine of the reversal.

**Level 2 — cost self-hedging.** When Farmer removes the most expensive items, it inadvertently reduces Green’s future spending. Each Farmer removal replaces one of Green’s upcoming purchases (if Green wanted the item) with a cheaper substitute, or leaves Green unaffected (if Green did not want the item). Either way, Farmer’s interference can never *increase* Green’s future cost. The rival’s aggression makes value-first more affordable, not less.

**Level 3 — absorbing value interference through the fluid limit.** Farmer’s deletions do shift Green’s ecological opportunities — an item Green would have picked next may be gone — but the fluid-limit value path accounts for this automatically. Because the proof tracks Green’s *actual* per-turn ecological value (the conditional quantile of the survivor pool at each moment, derived from the exact dynamic survivor law), Farmer’s interference is already priced into the trajectory. There is no separate “damage correction” to bound.

**Level 4 — budget self-hedging.** Under budget parity, both agents have enough money to purchase nearly everything. The stopping share for value-first is determined by budget accounting, not by strategic cost management. The late-game dynamics (how many extra turns Green gets after Farmer exhausts) therefore reduce to a simple counting argument rather than a fragile cost-path calculation.

All four levels reinforce the same conclusion: the adversarial structure rewards directness (buying the best ecology) and punishes cleverness (optimising the ratio), because the rival’s behaviour is entangled with the focal agent’s cost structure in a way that the static knapsack problem ignores.

**The baseline symmetry is self-hedging in its purest form.** When  $\rho = 0$  and marginals are identical, Level 1 (value self-hedging) and Level 3 (absorbing value interference) collapse

together into a clean symmetry: Green’s picks are orthogonal to Farmer’s cost ordering, so Farmer’s removals do not destroy value-first’s options preferentially. The four-level machinery is not needed — the game has a closed-form solution and the gap is an algebraic identity. Section 3 develops this case; Section 4 returns to the four levels once symmetry breaks.

### 3 The baseline theorem ( $\rho = 0$ ): the symmetry identity

Under independence with identical marginals, the game has a hidden symmetry that makes the entire proof collapse to a one-line identity for the asymptotic gap:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}[\text{PC}_N^{\text{ME}} - \text{PC}_N^{\text{MX}}] = \mu_E \phi - S_G(\tau) \approx 0.216135.$$

This section explains where the symmetry comes from, why the identity has a clean economic reading as “racing versus ceding,” and how the numerical closure works.

#### 3.1 The symmetric game

When  $\rho = 0$  and both coordinates are  $U[0.1, 10]$ , the two agents are shopping in *orthogonal* dimensions. Farmer ranks items by agricultural value  $A$ ; Green (under ME) ranks them by ecological value  $E$ . Under independence, knowing an item’s rank on one dimension tells you nothing about its rank on the other. Farmer’s max- $A$  pick is, from Green’s perspective, ecologically random; Green’s max- $E$  pick is, from Farmer’s perspective, cost-random. This is what we called “agricultural blindness” in earlier framings of the proof: value-first neither avoids nor prefers expensive items.

The consequence is symmetric market depletion. Each turn, Farmer removes the top- $A$  item from the survivor pool and Green removes the top- $E$  item. Under independence, the survivor pool after any number of these alternating removals is characterised by two cutoffs — a Farmer frontier on the  $A$ -dimension and a Green cutoff on the  $E$ -dimension — which evolve independently. In the fluid-limit (large market) coordinates  $u = 1 - \Phi(x)$  and  $v = 1 - \Phi(c)$  measuring the tail mass above each cutoff, the two frontiers satisfy identical equations of motion

starting from identical initial conditions. So  $u(s) = v(s)$  at every share  $s$ , and both follow the closed form

$$u(s) = v(s) = 1 - \sqrt{1 - 2s}.$$

This is symmetry as a property of the game, not an assumption of the proof. It is what agricultural blindness buys you.

### 3.2 The symmetry identity: racing versus ceding

The symmetry has an immediate economic consequence. At any share  $s$ , the items remaining in the pool have  $A$ -values uniform on  $[\ell, x(s)]$  and (independently)  $E$ -values uniform on  $[\ell, c(s)]$ . Farmer's max- $A$  pick has  $A$ -value near  $x(s)$  and an  $E$ -value drawn uniformly from  $[\ell, c(s)]$  — i.e., Farmer takes a generic item's worth of ecological value. Green's max- $E$  pick has  $E$ -value near  $c(s)$  and an  $A$ -value drawn uniformly from  $[\ell, x(s)]$  — Green pays a generic item's worth of agricultural cost. By the symmetry  $x(s) = c(s)$ , these two “generic” averages are equal:

$$\text{Farmer's per-turn } E = \frac{1}{2}(\ell + c(s)) = \frac{1}{2}(\ell + x(s)) = \text{Green's per-turn } A.$$

This is the *cross-damage symmetry* of the baseline game: the ecological value Farmer captures from ME's contested pool equals the agricultural cost Green incurs from Farmer's contested pool. Each agent damages the other on the dimension it doesn't care about, and under  $\rho = 0$  with identical marginals, they do so in exactly equal measure. The importance of this observation is not that it says something deep about competition in general — it depends on both independence and identical marginals in a way that does not extend beyond this specification — but that it gives us a clean reference point to which the correlated case (§4) can be compared. Positive correlation will break this symmetry, and everything complicated in the  $\rho = 0.3$  proof is machinery for handling that breakdown.

Integrating over the active phase: Farmer's cumulative ecological capture (through Farmer's exhaustion share  $\tau$ ) equals Green's cumulative agricultural spend  $S_G(\tau)$ .

Now combine this with budget accounting. After Farmer exhausts, Green shops alone and

(in the fluid limit) buys every remaining item. Green's total ecological haul is therefore

$$\text{PC}^{\text{ME}}/N \rightarrow \mu_E - S_G(\tau),$$

because Green gets everything except what Farmer took, and Farmer's ecological take equals  $S_G(\tau)$  by symmetry. Ratio-greedy, on the other hand, concedes every item with  $A$ -value above Farmer's budget-determined threshold  $q_F$ . Under independence with identical marginals that is exactly a  $\phi$ -fraction of the market, each with average ecological value  $\mu_E$ . So MX's cession to Farmer is worth  $\mu_E \phi$  in ecological terms:

$$\text{PC}^{\text{MX}}/N \rightarrow \mu_E(1 - \phi).$$

Subtracting gives the identity

$$\mu_E \phi - S_G(\tau).$$

The gap is the difference between what MX *cedes by design* — the expensive third of the market, worth  $\mu_E \phi$  in ecology under independence — and what ME *loses by racing* — the cumulative agricultural cost  $S_G(\tau)$  it pays by competing head-to-head with Farmer. In one phrase: *ratio-greedy retreats; value-first races; the race is worth the cost.*

This is self-hedging in its cleanest form. Level 1 is absorbed into the cross-damage symmetry itself (Farmer can damage value-first no more than Green damages Farmer). Level 2 is trivially operative but not load-bearing (there's no cost pressure to hedge against under independence). Level 3 is absent (no value interference to absorb). Level 4 shows up in the  $\kappa = 1 - \tau$  identity, which says Green buys everything Farmer didn't.

### 3.3 Why Level 1 self-hedging is the pure form at $\rho = 0$

The striking feature of the baseline identity is that it gives the *exact* asymptotic gap, not a lower bound. Earlier proofs of the same theorem produced conservative bounds (margin  $\geq 0.168$  in one architecture,  $\geq 0.046$  in another) because they bounded what Green might lose from Farmer's interference and then closed with explicit calculation. The symmetry argument does not need any such bound: agricultural blindness makes Farmer and Green's per-turn captures distributionally identical, and the identity falls out of bookkeeping.

Economically, this tells us something about what the self-hedging mechanism looks like in its simplest form. Under independence with identical marginals, the two agents' cross-damage to each other is symmetric — Farmer's ecological capture equals Green's agricultural spend. So the relative advantage of picking directly on the objective dimension (ME) versus picking on a ratio (MX) cannot come from Farmer being less damaging to one than the other. It must come from elsewhere: from the fact that ratio-greedy voluntarily *cedes* the upper- $A$  fraction of the market by design. That ceded portion is worth  $\mu_E \phi$  in ecological terms under independence, and the race value-first pays to compete in the remaining pool is only  $S_G(\tau)$ . The gap is the difference.

A useful way to remember the baseline result is this: at  $\rho = 0$ , competition is a wash (cross-damage is symmetric), so the advantage of value-first comes entirely from ratio-greedy's own choice to retreat. At  $\rho > 0$ , this symmetry breaks, and the story acquires the richer dynamics developed in §4.

### 3.4 Numerical closure

The identity  $\mu_E \phi - S_G(\tau)$  involves two constants:

$\phi$ , **the MX Farmer-exhaustion share**. Defined by  $\int_{1-\phi}^1 Q_A(v) dv = \mu_A/2$ , which under the baseline uniform law reduces to the quadratic  $9.9 \phi^2 - 20 \phi + 5.05 = 0$ . The root in  $(0, 1/2)$  is  $\phi = (10 - \sqrt{50.005})/9.9 \approx 0.295816 \dots$

$\tau$ , **the ME Farmer-exhaustion share**. Defined by  $S_F(\tau) = \mu_A/2 = 2.525$ , where  $S_F(s) = 0.1 s + 3.3(1 - (1 - 2s)^{3/2})$ . Since  $S_F$  is strictly increasing on  $[0, 1/2]$ , bisection on the bracket  $[0.304718, 0.304719]$  brackets  $\tau$  to arbitrary precision.

Evaluating  $\mu_E \phi = 5.05 \times 0.295816 \dots = 1.493871 \dots$  and  $S_G(\tau) \in [1.277734, 1.277738]$  with  $S_G(s) = 0.1 s + 1.65(1 - (1 - 2s)^{3/2})$  gives

$$\mu_E \phi - S_G(\tau) \in [0.216133, 0.216137] = 0.21613 \dots > 0.$$

An 80-digit interval-arithmetic certificate (archived with the paper's replication materials) tightens this to  $[0.216134, 0.216136]$ . This is the exact asymptotic gap.



The visual shape of the argument is shown in Figure 1: value-first builds its lead rapidly in the contested phase (while Farmer is still shopping), and although ratio-greedy recovers partially in the post-Farmer phase and purchases for slightly longer, the early lead is never erased.

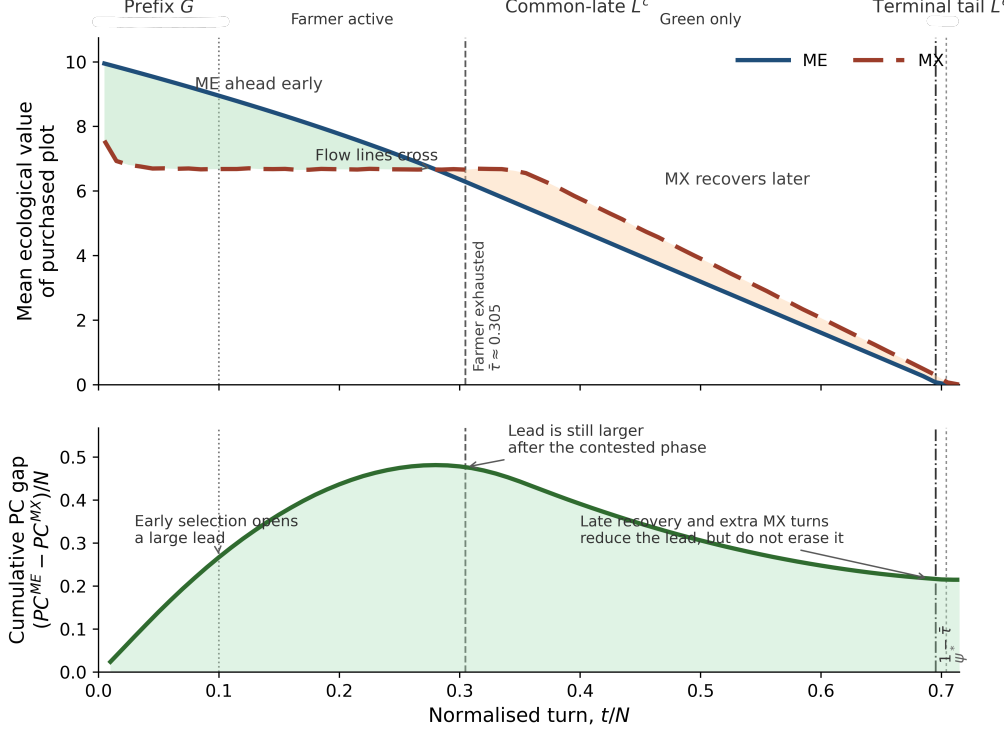


Figure 1: Visual anatomy of the baseline asymptotic dominance. Top panel: average ecological value of the plot purchased at each normalised turn under Max Environment and Max Efficiency in the baseline Budget World benchmark. The curves cross, so the theorem is not a pointwise-dominance result: Max Efficiency recovers later and purchases for slightly longer. Bottom panel: cumulative Purchased Conservation gap,  $(PC^{ME} - PC^{MX})/N$ . The contested phase (during which the Farmer is still shopping) creates a large advantage for Max Environment that the post-Farmer phase cannot undo. Vertical markers: prefix share  $m = 0.10$ , Farmer exhaustion  $\bar{\tau} \approx 0.305$  in the fluid limit. Baseline  $\rho = 0$ ,  $N = 2000$ , 3000 replications.

## 4 The correlated extension ( $\rho = 0.3$ ): when symmetry breaks

Under positive correlation the cross-damage symmetry between Farmer's and Green's picks is broken, and with it the clean baseline identity. Correlation is the empirically relevant case: on the Bolivian landscape, which motivates the paper, the estimated correlation between ecological and agricultural value indices is  $\rho \approx 0.29$ .

The clearest way to see what changes is in terms of the two cross-damage quantities that

were equal at  $\rho = 0$ . Farmer’s max- $A$  pick now disproportionately coincides with high- $E$  items, so Farmer’s per-turn ecological capture *exceeds* a generic item’s  $E$ -value — Farmer damages Green more than at baseline. Symmetrically, Green’s max- $E$  pick now correlates with high  $A$ , so Green’s per-turn agricultural cost also exceeds its baseline value — Green damages Farmer’s budget more. But these two effects do not grow at the same rate as  $\rho$  rises, and their asymmetry is exactly what the  $\rho = 0.3$  proof architecture is built to handle. The four levels of self-hedging introduced in §2 are now each doing specific work to rescue the reversal from this broken symmetry.

Positive correlation also *amplifies* the economic motivation for the reversal, even as it complicates the proof. The prefix wedge grows from +0.219 at baseline to +0.309 at  $\rho = 0.3$ : ratio-greedy’s corridor of high- $E/A$  items becomes even more ecologically mediocre under correlation (because high-ratio items are increasingly those with anomalously low  $A$  relative to their  $E$ ). So the  $\rho = 0.3$  theorem operates in a regime where the economic story has gotten stronger but the proof has gotten harder.

#### 4.1 Why correlation amplifies the reversal: the two-force story

The natural prior is that positive correlation should *help* ratio-greedy. When high- $E$  sites also have high  $A$ , a “best bang for the buck” rule seems better suited to finding hidden gems — sites with above-average ecology at below-average cost.

The theory shows the opposite, but not uniformly. The simulated gap  $\Delta(\rho)/N = [\text{PC}_N^{\text{ME}} - \text{PC}_N^{\text{MX}}]/N$  is hump-shaped in  $\rho$ : rising from  $\rho = 0$  through intermediate values, peaking near  $\rho \approx 0.7$ , and declining sharply as  $\rho \rightarrow 1$ . The gap remains positive across the entire feasible range, but its magnitude varies nontrivially.

This hump can be read as the result of two rising forces with opposite sign on the gap.

**The selection force.** As  $\rho$  rises, high-ratio items become increasingly the items with anomalously low  $A$  relative to their  $E$ : cheap-but-mediocre sites developers aren’t contesting. Ratio-greedy is drawn toward them more strongly; value-first avoids them by construction. The *prefix wedge*  $W(\rho)$  — the expected ecological gap between the top- $E$  and top- $E/A$  portfolios under the DGP at correlation  $\rho$  — therefore rises monotonically in  $\rho$ . This is Level 1 self-hedging

becoming more pronounced: Farmer’s max- $A$  aggression now destroys genuinely valuable options for ratio-greedy while leaving value-first’s already-secured portfolio intact. The selection force pushes the gap upward.

**The cross-damage force.** At  $\rho = 0$ , Farmer’s max- $A$  pick has an ecological value distributed like a generic item in the survivor pool. As  $\rho$  rises, Farmer’s max- $A$  pick correlates with high  $E$ , so Farmer’s per-turn ecological capture rises above the generic average. At  $\rho = 1$ , Farmer’s max- $A$  item *is* the max- $E$  item in the pool: Farmer captures the ecological maximum of the survivor pool on every turn, which is much larger than a generic item’s  $E$ . The cross-damage force, which we can measure at a representative mid-active-phase share by  $F_E(\rho)$  (the expected  $E$ -value of the current max- $A$  item at share  $s = 0.15$ ), therefore rises with  $\rho$  and *accelerates* as  $\rho \rightarrow 1$ . This force pushes the gap downward.

**The hump is the balance.** Both forces rise with  $\rho$ , but  $W(\rho)$  rises fairly evenly while  $F_E(\rho)$  is gentle at low  $\rho$  and accelerates toward its  $\rho \rightarrow 1$  limit, where Farmer captures the pool’s ecological maximum on every turn. The simulated gap rises with  $\rho$  while the selection force dominates, reaches its peak near  $\rho \approx 0.7$  where the two rising forces balance at the margin, and declines as the accelerating cross-damage force overtakes. At  $\rho = 0.99$ , the gap is still positive but has fallen by roughly half from its peak. Figure 2 shows the shape comparison directly.

It is worth stating what this decomposition does and does not claim.  $W(\rho)$  and  $F_E(\rho)$  are proxy quantities operating over different phases of the game; their difference does *not* equal  $\Delta(\rho)/N$  pointwise, and we should not expect a three-term additive accounting. What the two quantities establish is that the qualitative hump arises from the tension between these two rising forces, with the cross-damage force coming from the breakdown of the  $\rho = 0$  symmetry that governs the baseline theorem.

Theorem 4.2 formally proves the reversal at  $\rho = 0.3$ , which is well inside the rising phase of the hump. The declining phase beyond  $\rho \approx 0.7$  is known only through simulation; formal extension would require a different proof architecture because the fluid-limit ODE’s closure conditions fail as cross-damage becomes maximal. The policy implication is direct: cost-effectiveness mandates are most counterproductive precisely where ecological and agricultural values are moderately but not perfectly aligned — the empirically common case, and the one where the theorem’s quantitative guarantee is strongest.

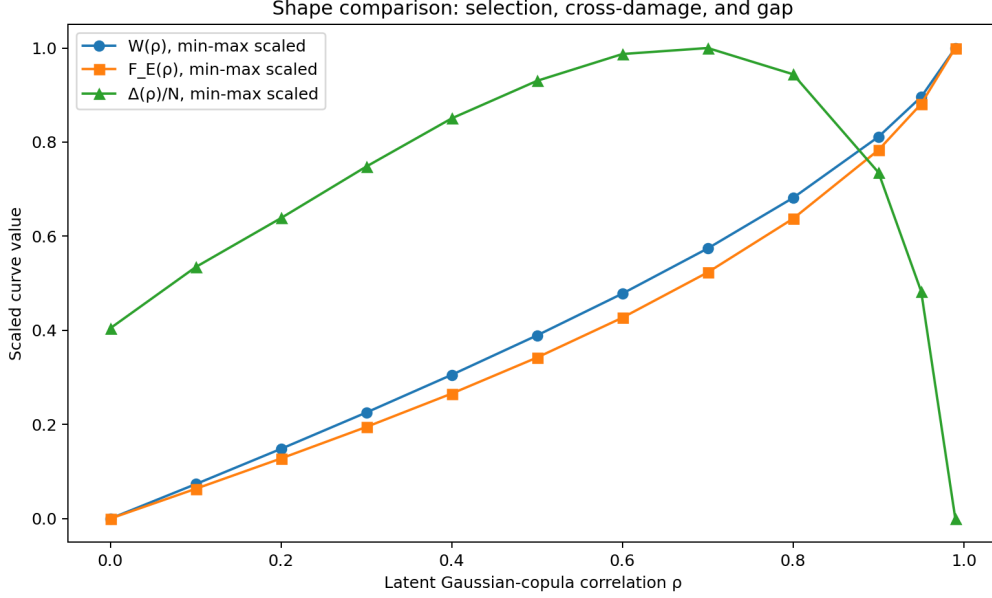


Figure 2: Two-force diagnostic for the hump-shaped  $\rho$ -gap. All three curves are min-max scaled to  $[0, 1]$  for shape comparison. The selection proxy  $W(\rho)$  (circles) rises approximately linearly in  $\rho$ , reflecting the growing quality gap between the top- $E$  and top- $E/A$  portfolios. The cross-damage proxy  $F_E(\rho)$  (squares) is gentler at low  $\rho$  and accelerates toward its  $\rho \rightarrow 1$  limit. The simulated  $\Delta(\rho)/N$  (triangles) rises while selection amplification dominates, peaks near  $\rho \approx 0.7$ , and declines as the accelerating cross-damage force overtakes. Simulation:  $N = 2000$ , 1000 replications per  $\rho$ , budget parity, Farmer moves first.

## 4.2 Agricultural blindness is lost: the monotone argmax

At  $\rho = 0.3$ , value-first is no longer agriculturally blind. When Green picks the best remaining  $E$ -value, correlation biases that pick toward high  $A$  as well. Green is now disproportionately selecting expensive items, and its picks accelerate the depletion of Farmer's frontier.

A monotone-argmax lemma formalises this. Among current survivors, items with higher  $A$  are stochastically more likely to have the highest  $E$  (because the conditional distribution of  $E$  given  $A$  is monotone in  $A$  under the Gaussian copula). So Green's argmax- $E$  pick is biased toward the top of the  $A$ -distribution, and the probability that Green's pick lies in the top- $k$   $A$ -block is at least  $k$  divided by the current survivor count.

This is where the proof architecture diverges from the baseline. Without symmetry, the frontier dynamics are coupled: Green's cutoff depends on Farmer's, Farmer's depends on Green's, and both must be tracked through the fluid-limit ODE. No closed form is available.

### 4.3 The cost host and the fluid-limit value path

Under correlation, conditioning on low  $A$  (cheap items) shifts the distribution of  $E$  downward: a pool of cheap items is, on average, ecologically poor. This creates a tension in the proof: Green’s cost certification wants to charge Green’s spending to a small, cheap “cost host” where upper bounds are tight, but Green’s value certification wants to credit Green’s ecological capture from a richer pool.

The resolution is to use two different deterministic objects for the two sides of the proof:

**Cost side: the cost host.** A small frontier-based pool  $\{A \leq Q_A(\sqrt{1-2\phi})\}$  on which Green’s agricultural spending can be bounded tightly. Level 2 self-hedging (below) operates on this host: Farmer’s max- $A$  deletions can only reduce Green’s spending on any host pool.

**Value side: the global fluid-limit value path.** Instead of bounding what Green might *lose* from Farmer’s interference (which an earlier architecture attempted, and which was too conservative to close the theorem), the current proof tracks what Green *actually gets* at each turn via the fluid-limit ODE. The ecological value integral  $\int_m^{\kappa_{\text{low}}} e(c(s)) ds$  accumulates Green’s asymptotic per-turn ecological value through the certified affordable horizon  $\kappa_{\text{low}} = 0.620816$ .

This two-host architecture is unique to the correlated case. At  $\rho = 0$ , one pool suffices because agricultural blindness makes the ecological distribution independent of any  $A$ -conditioning.

### 4.4 Level 2: cost self-hedging via the maxA-topY lemma

The cost side of the proof rests on one game-specific monotonicity property: when Farmer removes the most expensive item from a pool, Green’s future spending on that pool cannot increase. If the removed item was one Green intended to buy (because it also had high  $E$ ), Green substitutes a cheaper alternative (cheaper because the removed item was the global max- $A$ ). If the removed item was one Green did not want, Green is unaffected.

This is Level 2 self-hedging, and it is a *pathwise* (deterministic) result that holds for any correlation, any distribution, and any history of prior deletions — it requires only that Farmer

always removes the current max- $A$  item. The proof is a two-line case analysis on a finite set.

Economically, this tells us that the naïve structure of the Farmer’s strategy is what makes the cost certification work. If Farmer were a strategic optimiser — removing items to maximise damage to Green rather than to maximise its own near-term  $A$ -capture — the cost-side self-hedging would fail, because Farmer could target items whose removal forces Green into expensive substitutes.

The scope of the naïve assumption deserves emphasis. Most economic theory assumes firms are rational profit-maximisers, so a first reaction to “naïve max- $A$  Farmer” might be that the theorem covers an artificially narrow case. It does not. In the adversarial procurement literature, “developer pressure” is almost universally modelled as myopic: developers buy the site that delivers the highest immediate development value, not the site that most damages a rival conservation agency. This is both empirically realistic (real developers are not optimising a game against a conservation rival) and institutionally standard (development-value maximisation is how profit maximisation translates into this context). The theorem applies to this very wide class. Strategic Farmers — developers behaving as game-theoretic optimisers in an adversarial procurement setting — are an interesting further research question that the current proof architecture does not address.

#### 4.5 Level 3: tracking what Green actually gets

At every moment in the game, the survivor pool has a definite statistical structure: it is an iid sample from the doubly truncated Gaussian copula law, indexed by the two moving cutoffs — the Farmer frontier  $x$  and the Green cutoff  $c$ . This is the content of the *exact dynamic survivor law*, which underwrites the entire fluid-limit architecture.

Green’s per-turn ecological value is a deterministic function of the cutoffs:  $e(c(s))$ . As the game progresses, both cutoffs move downward at rates determined by both cutoffs jointly (because Green’s picks are biased toward expensive items under correlation, and Farmer’s picks are ecologically non-random). This coupled evolution is captured exactly by a four-dimensional ODE for the state  $(x, c, b_F, b_G)$  — the two cutoffs and the two remaining budgets.

The ODE is the centrepiece of the correlated proof. Under independence the ODE has a

closed-form solution and we do not need to solve it numerically; under correlation we do. A validated interval-Euler integration establishes that Green’s cumulative ecological value through  $\kappa_{\text{low}} = 0.620816$  is at least 2.5785, against an MX threshold of at most 2.3642. The certified margin is  $\geq 0.214$ .

This is Level 3 self-hedging: the fluid limit *absorbs* Farmer’s value interference automatically. The proof does not need to bound the ecological damage Farmer inflicts on Green; it needs only to integrate Green’s actual per-turn value. Green’s ecological captures are priced into the trajectory through the joint evolution of the cutoffs.

#### 4.6 Level 4: budget self-hedging and the stopping bound

Budget parity guarantees that Green has enough money to buy nearly everything, in the following sense. Green’s budget is  $\mu_A/2$ , and Green pays less per turn than Farmer does (because Green shops from below the frontier while Farmer shops at the frontier). A direct integration of Green’s budget ODE, together with the affordability bootstrap (which establishes that Green plays exact max- $E$  throughout the relevant horizon), shows that Green remains affordable through  $\kappa_{\text{low}} = 0.620816$ .

After Farmer exhausts, Green shops alone against a frozen frontier and continues to accumulate ecological value until either its budget runs out or the market is empty. In the finite game at most  $O(1)$  items remain unclaimed, so Green buys effectively everything below the frontier.

The importance of this level is that it prevents the reversal from unravelling through budget pressure. A different budget structure — one where Green exhausts before buying enough items to accumulate the asymptotic ecological value — would break the theorem. Under parity, budget self-hedging works: the budget accounting bounds the stopping share by a simple identity rather than by a fragile cost-path calculation.

#### 4.7 Numerical closure at $\rho = 0.3$

Putting the four levels together, the  $\rho = 0.3$  theorem reduces to a single numerical object: the value of the integral  $\int_m^{\kappa_{\text{low}}} e(c(s)) ds$  along the fluid-limit ODE trajectory. The validated interval-Euler integration gives

$$\int_m^{\kappa_{\text{low}}} e(c(s)) ds \geq 2.5785, \quad \Theta_{0.3} \leq 2.3642,$$

so the certified asymptotic gap is  $\geq 0.214$ . Simulations at moderate  $N$  suggest the true asymptotic gap is closer to 0.29; the certificate’s conservativeness reflects the interval-arithmetic padding and the fact that the integration stops at  $\kappa_{\text{low}}$  rather than at full budget exhaustion.

### 5 When does the reversal hold? Ecological heterogeneity

The self-hedging mechanism requires one substantive ingredient: enough ecological heterogeneity for the top- $E$  portfolio and the top- $E/A$  portfolio to actually differ. When ecological values are nearly homogeneous — every site has roughly the same  $E$  — value-first and ratio-greedy select similar portfolios and the reversal disappears. The prefix wedge shrinks, Level 1 stops doing work, and the elegant architecture of the proof loses its driver.

The paper’s *ecological selection premium* (ESP) diagnostic operationalises this requirement. ESP is defined as the percentage by which the average ecological value of the top- $E$  portfolio exceeds that of the top- $E/A$  portfolio on any given landscape:

$$\text{ESP} = \frac{\bar{E}_{\text{top-}E} - \bar{E}_{\text{top-}E/A}}{\mu_E}.$$

Compressed- $E$  simulations indicate that the reversal holds empirically whenever ESP exceeds roughly 12.5%, while the current proof architecture requires approximately 15%. On the paper’s baseline grids and on the Bolivian landscape, ESP exceeds 30% — well above either threshold. The diagnostic is straightforward to compute from any landscape dataset and offers practitioners a check on whether the mechanism is operative in their setting.



The policy implication is direct: cost-effectiveness mandates are most counterproductive precisely where ecological and agricultural values are positively aligned and ecological heterogeneity is large. The Bolivian landscape is the empirically common case in both respects.

## 6 What the proofs show, and a note on robustness

The two theorems are proved for a specific data-generating process (uniform marginals on  $[0.1, 10]$ , Gaussian copula) at two correlation values. They do NOT prove:

- Universal dominance for all finite instances. Constructive counterexamples on very small grids ( $2 \times 2$ ,  $2 \times 3$ ) show that with tight budgets a single expensive asset can exhaust Green's budget, allowing ratio-greedy to outperform value-first. These pathological configurations are confined to grids where the budget concentration ratio  $\kappa(N) = \max_i A_i / B_G$  is  $O(1)$  rather than shrinking like  $O_p(1/N)$ . By  $3 \times 3$  the pathology is already gone, and the Bolivian case study ( $N = 390$ ,  $\kappa \approx 0.01$ ) sits deep in the region where self-hedging dominates.
- Dominance at all correlation levels. Only  $\rho = 0$  and  $\rho = 0.3$  carry formal proofs. Simulations suggest the reversal holds across the entire feasible range  $\rho \in [0, 0.995]$ , with a hump-shaped profile peaking near  $\rho \approx 0.7$ .
- Dominance under strategic Farmers. The theorems use naïve max- $A$  Farmers. Strategic Farmers — those optimising over time rather than myopically — are studied only in simulations. The Level 2 self-hedging result (Farmer's deletions cannot raise Green's costs) depends essentially on Farmer being naïve, so the reversal is not automatic in the strategic case.
- Dominance under ecological homogeneity. When  $E$ -values are nearly homogeneous, ESP is small and the prefix wedge disappears. The theorems' conditions fail here, and simulation evidence suggests the reversal itself fails.

**A note on proof robustness.** The asymptotic dominance result at  $\rho = 0$  has been proved three times using substantially different architectures. The primary proof in [Weinhold and](#)

Andersen (2026) uses the E–A symmetry identity developed in Section 3 above and obtains the asymptotic gap as a closed-form identity ( $\mu_E \phi - S_G(\tau) \approx 0.216$ ). An earlier fluid-limit proof certified a conservative lower bound of 0.168 via an explicit value integral against the MX threshold. A still earlier worst-case deletion proof certified a further-conservative lower bound of 0.046 by bounding what value-first might lose from Farmer’s interference rather than tracking what it actually gets. All three reach the same qualitative conclusion — value-first strictly dominates ratio-greedy asymptotically — through independent analytical routes. That the result survives this much architectural change is a quiet form of robustness: the knapsack reversal is not an artefact of any particular proof technique. The two alternative proofs and their validated numerical certificates are available from the author upon request.

At  $\rho = 0.3$  the result rests on a single proof architecture (the fluid-limit ODE with validated numerical closure). A symmetry-based alternative is not available because the symmetry itself breaks under positive correlation; likewise the worst-case deletion architecture was tried and found to be too conservative to close the theorem (certified margin of approximately 0.003 against a simulation gap of 0.29). The fluid-limit architecture’s ability to track Green’s actual per-turn ecological value, rather than bound its losses, is what makes the correlated theorem tractable.

## References

Weinhold, D. and Andersen, L. (2026). Adversarial dynamics and the knapsack reversal: Insights and evidence from conservation siting. London School of Economics, working paper. Available at [https://github.com/dmweinhold/Latest-Version-Papers/raw/main/Adversarial\\_Procurement.pdf](https://github.com/dmweinhold/Latest-Version-Papers/raw/main/Adversarial_Procurement.pdf).